

Alexandra Macedo

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Education

California State University, Fullerton (CSUF)

M.S. in Physics (GPA: 3.94)

Expected December 2025

Fullerton, CA

University of California, San Diego (UCSD)

M.S. in Computer Science (specialization in Artificial Intelligence) (GPA: 3.6)

December 2020

San Diego, CA

University of California, Los Angeles (UCLA)

B.S. in Molecular, Cell, and Developmental Biology (GPA: 3.6)

March 2015

Los Angeles, CA

Projects

C++ Tensor Expression Templates | C++17, C++20, Blaze, template metaprogramming

2020 – present

- Developed expression templates for high-performance tensor arithmetic in Einstein notation for open-source numerical relativity code, SpECTRE (spectre-code.org) [9]
- Optimized expression runtime to be faster than or on par with loop implementations of tensor arithmetic [10]

Finite Element Mesh Improvements for Astrophysical Simulations | C++17, C++20

2024

- Improved the robustness of computational meshes to allow for highly dynamical black hole evolutions [8]
- Played a key role in getting binary black hole simulations to merge robustly and co-authored a scientific paper [2]

Modeling LIGO's Astronomical Range with Sensor Data | Python, scikit-learn, GWpy, GWDetChar

2018

- Performed feature selection to reduce dimensionality from ~500,000 to ~10 to diagnose noise sources for gravitational-wave observatories and co-authored a scientific paper [4] [12]
- Automated daily modeling and data visualization for the gravitational-wave scientific collaboration, LVK

Experience

Software Developer, Research Assistant

Nov 2017 – Sept 2018, Jun – Aug 2019, Jun 2020 – present

California State University, Fullerton

Fullerton, CA

- Contributed over 40K lines to SpECTRE, a task-based, massively parallel C++17 astrophysics code, and contributed to GWDetChar, a gravitational-wave detector characterization code
- Implemented a more efficient parallel resource allocation strategy on systems with hundreds of nodes, reducing runtime of equal mass binary black hole evolutions by up to 75%
- Reduced compile memory and time of heavily templated executables by up to 20% and 18%, respectively

Teaching Assistant

Sept 2024 – present

California State University, Fullerton

Fullerton, CA

- Independently run a lower division electromagnetism laboratory course of 10-20 students

Teaching Assistant

Sept 2018 – Jun 2019, Sept 2019 – Jun 2020

University of California, San Diego

San Diego, CA

- Instructed and developed weekly bespoke software for automatically grading 300-600 students in Java Programming; Data Structures; and Computer Organization and Systems Programming
- Co-authored two online interactive programming textbooks in continued use in undergraduate courses [6] [7]
- Mentored 30 graduate computer science teaching assistants, leading weekly group educational practica

Skills

Languages: C++11/14/17/20, C, Python, Java, Bash, LaTeX

Technologies: Git, GitHub, Linux/UNIX shell, STL, Blaze, Brigand, CMake, Make, VS Code, Vim, Docker, Slurm

Concepts: Template metaprogramming, test driven development, continuous integration, code review, pull requests, benchmarking, profiling, optimization, high performance computing, computational astrophysics, physics simulation

Communication: Teaching, mentoring, public speaking, technical writing

Awards

UCSD Computer Science and Engineering Masters Award for Excellence in Teaching (2020)

UCSD GradWIC Grace Hopper Conference Scholarship Fund (2020)

UCSD Golden State Teaching Scholarship (2018)

Volunteerism and Leadership

- Mentor Teaching Assistant (UCSD, 2019-2020)
- Graduate Mentor (UCSD JUMP, 2019-2020)
- English and American Culture Instructor (UCLA DCISS, 2016-2018)
- President, Member (UCLA Drug Outreach and Education Program, 2013-2015)
- Community Service Commission Liason, Coach (Special Olympics at UCLA, 2013-2015)

Affiliations and Memberships

American Physical Society (2021, 2024–present)
Simulating eXtreme Spacetimes Collaboration (2019 – present)
Nicholas and Lee Begovich Center for Gravitational-wave Physics and Astronomy (2017 – present)
LIGO Scientific Collaboration (now LIGO-Virgo-KAGRA Collaboration) (2017 – 2019)
UCSD Graduate Women in Computing (2018 – 2020)

Publications

Submitted

- [1] Nikolas Wittek, ..., **Alexandra Macedo**, et al. “Relieving scale disparity in binary black hole simulations.” [\[arXiv\]](#)

Peer reviewed

- [2] Geoffrey Lovelace, ..., **Alexandra Macedo**, et al. “Simulating binary black hole mergers using discontinuous Galerkin methods.” *Classical and Quantum Gravity*, 2025. [**CQG**] [\[arXiv\]](#)
- [3] Nils Deppe, ..., **Alexandra Macedo**, et al. “Simulating magnetized neutron stars with discontinuous Galerkin methods.” *Phys. Rev. D*, 105(12):123031, 2022. [**PRD**] [\[arXiv\]](#)
- [4] Marissa Walker, ..., **Alexandra Macedo**, et al. “Identifying correlations between LIGO’s astronomical range and auxiliary sensors using lasso regression.” *Classical and Quantum Gravity*, 2018. [**CQG**] [\[arXiv\]](#)
- [5] Sander Valstar, ..., **Alexandra Macedo**, et al. “Faculty Views on the Goals of an Undergraduate CS Education and the Academia-Industry Gap.” *Association for Computing Machinery*, 2020. [**ACM**]

Educational

- [6] Sophia Krause-Levy, ..., **Alexandra Macedo**, et al. “Computer Organization and Systems Programming (C and ARM).” *Stepik*, 2020. [**Stepik**]
- [7] Joseph Gibbs Politz, **Alexandra Macedo**, et al. “Intro to Programming in Python then Java using Media Computation.” *Stepik*, 2020. [**Stepik**]

Conference Presentations

Future

- [8] **Alexandra Macedo**, Marceline Bonilla, et al. “Improvements to SpECTRE’s Computational Domain for Binary Black Hole Simulations.” American Physical Society April Meeting, 2025.

Past

- [9] **Alexandra Macedo** and Nils Deppe. “Writing Tensor Equations in SpECTRE.” Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM, 2024.
- [10] **Alexandra Macedo** and Nils Deppe. “Design, optimization, and updated status of SpECTRE’s C++ tensor expression interface.” American Physical Society April Meeting, 2024.
- [11] **Alexandra Macedo** and Nils Deppe. “SpECTRE’s C++ tensor expression interface.” American Physical Society April Meeting, 2021.
- [12] **Alexandra Macedo**, Marissa Walker, et al. “Identifying correlations between LIGO’s astronomical range and auxiliary sensors using LASSO regression.” 34th Pacific Coast Gravity Meeting, 2018.

References

Nils Deppe

Assistant Professor of Physics

Cornell University

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Relation: C++ Tensor Expressions project mentor, lead SpECTRE developer

Geoffrey Lovelace

Professor of Physics

California State University, Fullerton

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215-740-9176

Relation: Current research advisor

Joshua Smith

Professor of Physics, GWPAC Director

California State University, Fullerton

josmith@fullerton.edu

657-278-3716

Relation: Past research advisor, current research center director, graduate mechanics course professor

Gina Passante

Associate Professor of Physics

California State University, Fullerton

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Relation: Graduate quantum mechanics course professor

Joseph Gibbs Politz

Associate Teaching Professor of Computer Science

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858-534-8872

Relation: Past teaching assistantship professor

Leo Porter

Professor of Computer Science

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858-534-8872

Relation: Past teaching assistantship professor